



Exam Questions

What is the main reason for reducing an aircraft's weight?

- a) Improve passenger comfort
- b) Increase aerodynamic drag
- c) Reduce fuel consumption and costs
- d) Improve takeoff speed

What is an aircraft structure primarily responsible for?

- a) Controlling flight dynamics
- b) Carrying loads and withstanding forces
- c) Reducing drag
- d) Generating lift

Exam Questions

What is an airframe?

- a) The entire aircraft including engines and avionics
- b) The basic structure of an aircraft without equipment or interior installations
- c) The main wings and fuselage combined
- d) The control surfaces only

What type of energy is converted into thrust in an aircraft?

- a) Kinetic energy
- b) Thermal energy
- c) Electrical energy
- d) Potential energy

Exam Questions

Lower weight in an aircraft leads to:

- a) Increased emissions
- b) Higher fuel consumption
- c) Reduced operational costs
- d) More structural failure

What is the primary structure in an aircraft?

- a) A non-load-bearing component
- b) A structure that carries critical loads and failure is catastrophic
- c) A structure that only affects aerodynamics
- d) A replaceable part that does not affect the overall strength

Exam Questions

What is a secondary structure in an aircraft?

- a) A structure that does not contribute to strength
- b) A structure that carries loads but its failure is not catastrophic
- c) A replaceable structural component
- d) A component that only supports passenger safety

Which of the following is NOT part of the fuselage structure?

- a) Stringers
- b) Frames
- c) Ribs
- d) Longerons

Exam Questions

What is the main function of bulkheads in an aircraft fuselage?

- a) Improve aerodynamic efficiency
- b) Support avionics installation
- c) Make the fuselage airtight and provide structural reinforcement
- d) Reduce drag at high speeds

What is the function of longerons in an aircraft?

- a) They provide longitudinal and bending stiffness
- b) They are used to store fuel
- c) They control wing movement
- d) They are used in the cockpit area only

Exam Questions

What is tension in aircraft structures?

- a) A force pulling an object apart
- b) A force pushing an object together
- c) A force that acts tangentially
- d) A force causing twisting motion

What type of force occurs when an object is being pushed together?

- a) Shear
- b) Compression
- c) Tension
- d) Torsion

Exam Questions

What is torsion in an aircraft structure?

- a) A force that causes bending
- b) A twisting moment along the longitudinal axis
- c) A force acting parallel to a surface
- d) A force that causes displacement

Which type of force occurs when equal and opposite forces act parallel to a surface?

- a) Bending
- b) Shear
- c) Compression
- d) Torsion

Exam Questions

Which type of loading is a combination of tension, compression, and shear?

- a) Bending
- b) Torsion
- c) Axial loading
- d) Buckling

A truss structure is made of:

- a) A single metal plate
- b) Bars, tubes, and wires
- c) Reinforced concrete
- d) A flexible membrane

Exam Questions

What is the main limitation of wires in a truss structure?

- a) They cannot withstand compression
- b) They add too much weight
- c) They are too expensive to manufacture
- d) They can only be used in small aircraft

Shell structures use sheet material mainly to:

- a) Carry aerodynamic loads
- b) Reduce weight
- c) Resist shear loads
- d) Increase temperature resistance

Exam Questions

What happens if the stress in a shell structure is too large?

- a) It flexes but remains stable
- b) It withstands compression effectively
- c) It deforms or buckles
- d) It becomes stronger over time

How can a shell structure be reinforced?

- a) Adding more shear loads
- b) Using stiffeners
- c) Increasing the aspect ratio
- d) Reducing the number of frames

Exam Questions

Beams in an aircraft separate:

- a) Normal and shear forces
- b) Lift and thrust forces
- c) Drag and pressure forces
- d) Engine and fuselage forces

A sandwich structure consists of:

- a) A metal core
- b) A low-density core and two face sheets
- c) Two trusses joined together
- d) Multiple thin beams

Exam Questions

What is a major drawback of sandwich structures?

- a) They absorb too much heat
- b) They are prone to moisture absorption
- c) They are too heavy
- d) They cannot carry loads

What is the main function of spars in a wing?

- a) Transfer aerodynamic loads to the fuselage
- b) Carry wing bending loads
- c) Reduce aircraft drag
- d) Store avionics

Exam Questions

How many spars do most aircraft wings have?

- a) 1
- b) 2
- c) 4
- d) 6

What does a torsional box do in a wing?

What is the function of wing ribs?

Exam Questions

What is a bulkhead used for?

Fuselage skins carry:

What is the main function of a payload fairing?

Answers

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- b) 2**
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- d) 6

What does a torsional box do in a wing?

Distributes torsional loads more efficiently

What is the function of wing ribs?

Maintain aerodynamic shape

Exam Questions

What is a bulkhead used for?

Provides pressurization and structural integrity

Fuselage skins carry:

Shear loads

What is the main function of a payload fairing?

Protects the payload from aerodynamic forces during launch